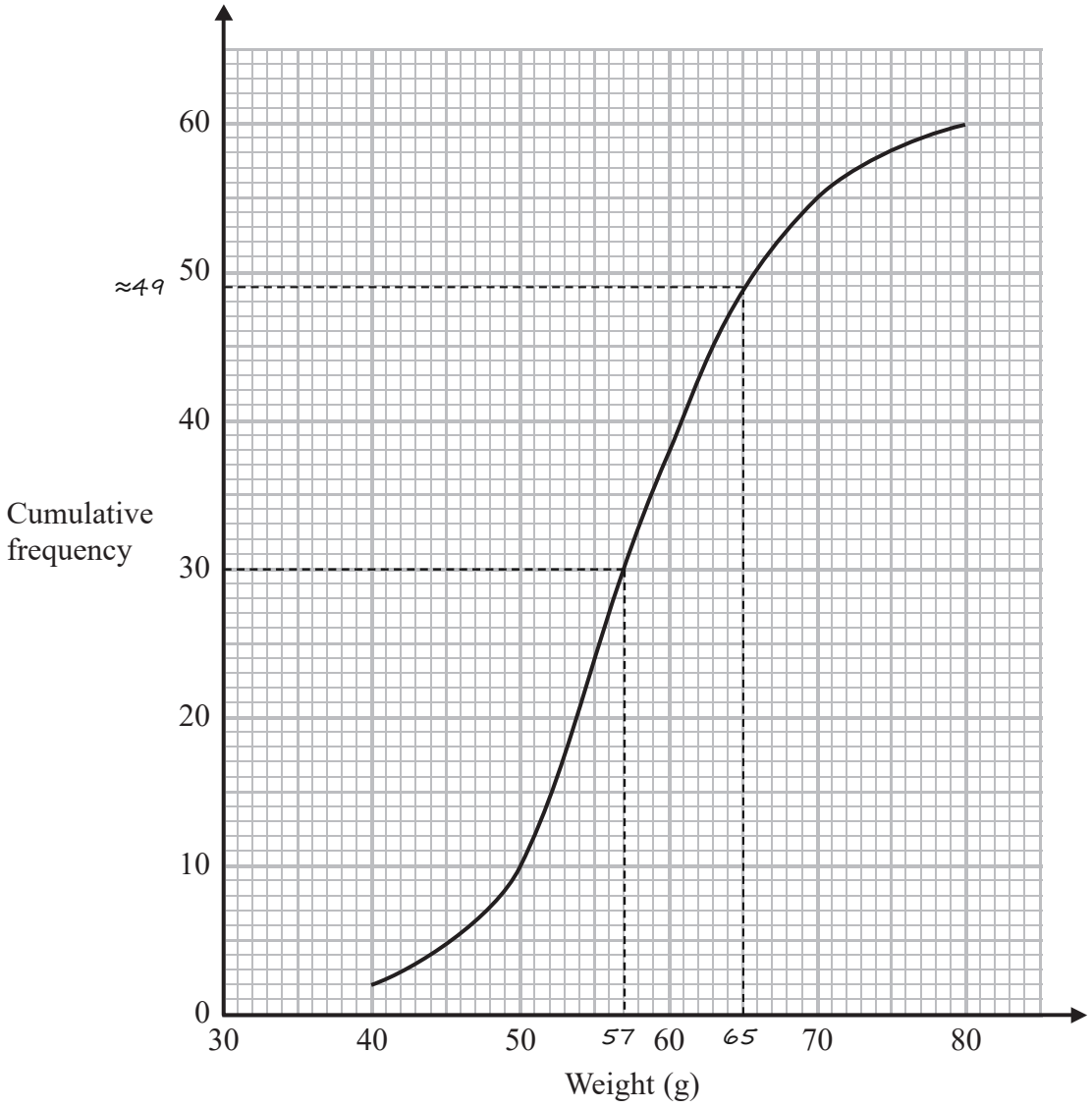
11 The cumulative frequency graph shows information about the weights of 60 potatoes.



(a) Use the graph to find an estimate for the median weight.

Half of 60 is 30. Read across at CF = 30, then down.

57 g

(1)

1 mark
Median estimate 57 g

Jamil says,

"80 − 40 = 40 so the range of the weights is 40 g."

(b) Is Jamil correct?
You must give a reason for your answer.

Not necessarily. The graph does not give the exact smallest or largest weight; for example, the smallest could be below 40 g.

1 mark
A range of exactly 40 g is not established

(1)

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(c) Show that less than 25% of the potatoes have a weight greater than 65 g.

From the graph, about 49 potatoes weigh 65 g or less.

1 mark
Read the cumulative frequency at 65 g

About 60 – 49 = 11 potatoes weigh more than 65 g.

25% of 60 = 15. Since 11 < 15, fewer than 25% weigh over 65 g.

1 mark
Compare with 15 potatoes, which is 25% of the sample

(2)

(Total for Question 11 is 4 marks)

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